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SUPERSEDING
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MILITARY SPECIFICATION

TRAILERS, LOWBED: FORKLIFT TRANSPORTER, 5450 AND 8150 KILOGRAM (6 AND 9 TON), MODIFIED COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three different commercial lowbed trailers, each with a 3660 millimeter (12 foot) long platform, beavertail, hinged loading ramps and a loading winch, all for loading and transporting single forklift trucks and other wheeled material handling equipment. Vehicles produced under this specification are commercial items that are required to be warranted by the manufacturer as specified in acquisition documents.

1.2 Classification. The trailer shall be one of the following sizes, as specified (see 6.2):

Size 6 - 5450 kilograms (kg) (6-ton) payload capacity, 4-wheel, tandem axle.
Size 9 - 8150 kg (9-ton) payload capacity, 6-wheel, tri-axle.
Size 9-W - 8150 kg (9-ton) payload capacity, 4-wheel, tandem axle, 2590 mm (102 inches) wide.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document or by letter.

AMSC N/A

FSC-2330

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODDIS), and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

	W-B-131	- Battery Storage: Vehicular, Ignition, Lighting and Starting.

MILITARY

*	MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
	MIL-H-3912	- Hardwood: Floorboards and Platforms, for Military Vehicles.
	MIL-H-46170	- Hydraulic Fluid, Rust Inhibited, Fire-Resistant, Synthetic Hydrocarbon Base.

STANDARDS

FEDERAL

	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

	MIL-STD-1223	- Non-Tactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
*	MIL-STD-1474	- Noise Limits for Army Material.
*	MIL-STD-1595	- Qualification of Aircraft, Missile and Aerospace Fusion Welders.
	MS 51336	- Lunette Coupler, Drawbar, Ring.
	MS 75020	- Connector, Plug, Electrical - 12 Contact, Intervehicular, 28 Volt, Waterproof.
	MS 75021	- Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 28-Volt, Waterproof.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

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* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

* DEPARTMENT OF DEFENSE
Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations (FMCSR).
Federal Motor Vehicle Safety Standards (FMVSS).

* (Application for copies of DoT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)
Code of Federal Regulations, 29 CFR.

(Application for copies of OSHA publications should reference the Code of Federal Regulations, 29 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of documents cited in the solicitation (see 6.2).

* AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
ASME Boiler and Pressure Vessel Code (BPVC)
Section IX - Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies of ASME publications should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

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* AMERICAN WELDING SOCIETY (AWS)

AWS B2.1	- Standard for Welding Procedure and Performance Qualifications.
AWS D1.1	- Structural Welding Code-Steel.

(Application for copies of AWS publications should be addressed to the American Welding Society, 2501 NW 7th Street, Miami, FL 31025.)

THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual.

(Application for copies of ETRTO publications should be addressed to the European Tyre and Rim Technical Organization, 32 Avenue Brugman, 1060 Brussels, Belgium).

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)
SAE Standards and Recommended Practices

J318	- Air Brake Gladhand Service (Control) and Emergency (Supply) Line Couplers - Trucks, Truck-Tractors, and Trailers (DOD Adopted).
J516	- Hydraulic Hose Fittings.
J517	- Hydraulic Hose.
* J537	- Storage Batteries.
J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
J682	- Rear Wheel Splash and Stone Throw Protection.
J697	- Safety Chain of Full Trailers or Converter Dollies.
J702	- Brake and Electrical Connection Locations - Truck-Tractor and Truck-Trailer (DOD Adopted).
J706	- Rating of Winches.
J844	- Nonmetallic Air Brake System Tubing (DOD Adopted).
J931	- Hydraulic Power Circuit Filtration.
J1067	- Seven Conductor Jacketed Cable for Truck and Trailer Connections.
J1292	- Automobile, Truck, Truck-tractor, Trailer, and Motor Coach Wiring.

(Application for copies of SAE publications should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

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THE TIRE AND RIM ASSOCIATION, INC.
Year Book.

* (Application for copies of Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this document takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.11, the vehicle, components, assemblies and accessories to be delivered under the contract shall be the manufacturer's standard or optional items, sold commercially to the public, which meet or exceed the requirements of this specification. All vehicle items shall be as represented and rated in the vehicle and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material, identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity prior to delivery of the items. The vehicle shall be a standard commercial trailer, of a model sold to a significant number of buyers other than the Government, in the course of normal business operations. The model furnished shall be not older than the manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate military service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

* 3.1.1.2 Marking and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223.

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* 3.1.1.3 Rustproofing. The vehicle shall be rustproofed in accordance with FED-STD-297.

* 3.1.1.4 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the trailer manufacturer's standard matching hardware.

3.1.1.5 Welders and welding. In addition to the requirements of 3.21.4, all welders employed in the fabrication of the trailer shall be certified, before any welding is accomplished, in accordance with MIL-STD-1595; the welding qualifications of the ASME BPVC, Section IX; or AWS B2.1 or D1.1. The certification that the welders have passed the qualifications test shall be on file at the contractor's facility and shall be available for review by the Government.

* 3.1.1.6 Lubrication chart. An illustrated or schematic diagram type lubrication chart shall be provided on the trailer. The chart shall direct attention to all lubrication fittings and specify the range and grade of lubricants required for critical temperatures. The chart shall be permanently attached to the trailer in a readily visible location. The chart shall be inscribed on a plate conforming to 3.1.1.11.

* 3.1.1.7 Hardwood. Wood platforms (see 3.17.1) shall be floored with apitong, white or red oak, or white ash as specified in MIL-H-3912. Apitong is allowed for the flooring only, provided the average weight shall be not less than 753 kilograms per cubic meter (kg/m^3) (47 pounds per cubic foot (pcf)), with none of the individual pieces weighing less than 673 kg/m^3 (42 pcf), with no more than 15 percent moisture content at time of weighing. Flooring shall be Government source inspected.

* 3.1.1.8 Wood treatment. Wood, including apitong, shall be treated with wood preservative in accordance with MIL-STD-1223.

3.1.1.9 24-volt lighting.

3.1.1.9.1 Independent 24-volt dc system. When specified (see 6.2), in addition to the standard 12-volt direct current (dc) electrical system (see 3.13 through 3.13.7), a 24-volt dc lighting system, independent of the 12-volt dc lighting system, shall be provided. Separate wiring and bulbs shall be provided for the 24-volt dc system. The system shall conform to 3.13. The 24-volt dc system shall also conform to the Federal Motor Carrier Safety Regulations referenced in 3.13.1, except the 24-volt dc stop and turn lamps shall be combined.

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3.1.1.9.2 Interconnected 24-volt dc system. When specified (see 6.2), capability of operating the normal 12-volt dc system from a towing vehicle equipped with a nominal 24-volt dc electrical system shall be provided.

3.1.1.9.3 Receptacle for 24-volt dc system. When either an independent or interconnected 24-volt dc system is specified, the front of the trailer shall also be equipped with a 12-contact receptacle and cover conforming to MS 75021, part number MS 75021-1. The receptacle shall be located in accordance with SAE J702. The 24-volt dc, 12-contact receptacle shall be connected to the lights as follows:

Contact B. Connect to left-hand turn signal and stop lamp (yellow)
Contact D. Connect to ground (white)
Contact E. Connect to clearance, side marker, identification, and tail lamps (black and brown)
Contact J. Connect to right-hand turn signal and stop lamp (green).
Contact L. Connect to ground (white).

The remaining contacts shall not be connected. Circuits B and J on tactical (military design) trucks are combination stop and turn indicator circuits. On the interconnected 24-volt dc system, the normal 12-volt dc turn signal lights will function both as turn signals and stop lights, and the normal 12-volt dc stop lights will not be operational when the trailer is connected to a towing vehicle with a 24-volt dc power supply. Because of this condition, the stop light (red) circuit is not connected to the 24-volt, 12-contact receptacle.

3.1.1.9.4 Resistors. When an interconnected 24-volt dc system is specified, the 12-contact receptacle shall be provided with resistance in each circuit to reduce the voltage of a tactical towing vehicle from a nominal 28 volts dc to 12 volts dc. Each circuit resistor shall be selected to reduce the voltage of the tactical truck to within the maximum rated voltage of the trailer electrical components. The resistor assembly shall be located in a protective housing and provided with adequate ventilation or a heat sink to prevent overheating and any heat damage to resistors, wiring, or adjacent components.

3.1.1.9.5 Twelve-conductor lighting cable. A 12-conductor lighting cable, for either the interconnected or independent lighting system, shall be furnished. The cable shall be not less than 910 mm (3 feet) longer than the distance from the 24-volt receptacle to the lunette. Both ends of the cable shall be equipped with connectors conforming to MS 75020-1 and MS 75020-2. Provisions shall be made to secure the cable, when not in use, to the drawbar or to the trailer body.

3.1.1.10 Lunette. The trailer shall be equipped with a lunette conforming to MS 51336 and 3.11.

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* 3.1.1.11 Information plates. The plates specified in 3.1.1.6, 3.4.3 and 3.19.4.4 shall conform to composition A (class 1 or 2) or composition C of MIL-P-514.

* 3.2 General design.

* 3.2.1 Federal Motor Vehicle Safety Standards. The vehicle shall comply with all Federal Motor Vehicle Safety Standards applicable to the type of trailer furnished and in effect on the date of manufacture.

* 3.2.2 OSHA and FMCSR regulations. The vehicle, with only furnished attachments, accessories and equipment, shall enable compliance by any user with all OSHA and FMCSR user regulations that are vehicle or vehicle equipment dependent. Regulations shall be those that are applicable to a commercial user of such a similar type, class and size of trailer for the same general use. See 6.1.

* 3.2.3 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be available to the appropriate engineering office of the procuring activity.

* 3.2.4 Dissimilar metals. All dissimilar metals used throughout the vehicle shall be insulated from one another to prevent galvanic or electrolytic action.

* 3.2.5 Prohibited materials. Asbestos materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries.

* 3.2.6 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.7 Lubrication. Lubrication means shall be provided for all parts of the trailer normally requiring lubrication. Where the use of high lubricating pressure will damage grease seals or other parts, fittings with pressure release shall be used.

* 3.2.8 Safety. All equipment or exposed portions of the equipment which are subject to extreme temperatures and inclement weather and all rotating or reciprocating parts which are of such a nature or so located as to become a hazard to operating personnel shall be insulated, fully enclosed, or properly guarded.

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3.2.9 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551.

3.3 Performance. The trailer shall evidence no part failure, deformation, permanent set or interference between parts when towed, both empty and when loaded as specified in 3.4:

(a) At speeds as great as 32 kilometers per hour (km/h) (20 miles per hour (mph)) over unimproved roads and reasonably hard uneven terrain;
(b) At speeds as great as 97 km/h (60 mph) over improved roads; and
(c) At speeds as great as 16 km/h (10 mph) on 10 percent side slopes on unimproved roads and reasonably hard uneven terrain.

3.3.1 Turning ability. The trailer shall be capable of moving through a 55 degree angle relative to a coupled towing vehicle, without cramping or interference between the trailer and the towing vehicle. This angle shall be measured at the center of the lunette and shall be the angle between the longitudinal axis of the trailer and the longitudinal axis of the towing vehicle. The ability of the trailer to meet this requirement shall be demonstrated actually or graphically with a 2590 mm (102 inch) wide towing vehicle having its pintle at the extreme rear of the towing vehicle. Interference will be considered to exist if, at the 55 degree angle, any portion of the trailer, 1295 mm (51 inches) or less from the lunette, is forward of the pintle.

* 3.3.2 Brake performance. The service brakes shall stop the truck-trailer combination, with the trailer loaded as specified in 3.4, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52. Requirements shall be met with the truck loaded to its full rated gross vehicle weight (GVWR) and gross combination weight rating (GCWR).

3.3.3 Tracking ability. The trailer shall conform to the tracking requirement of Federal Motor Carrier Safety Regulation 393.70(a).

3.4 Payload and distribution. Rated payload capacity with the payload evenly distributed over the loadspace, shall be not less than 5450 kg (12,000 pounds) for size 6 trailer and not less than 8150 kg (18,000 pounds) for sizes 9 and 9-W trailers. The trailer shall be capable of supporting concentrated loads encountered in transporting such material handling equipment as a forklift truck in lieu of an evenly distributed payload. A forklift truck shall be considered as weighing 4766 kg (10,500 pounds) for size 6 trailer and 7035 kg (15,500 pounds) for sizes 9 and 9-W trailers.

3.4.1 Gross weight distribution. Gross weight (trailer curb weight plus payload) distribution shall be 82 to 88 percent on the tires and 12 to 18 percent on the lunette. Weight distribution shall be determined with the rated payload distributed as specified in 3.4, with the platform level and the landing leg retracted.

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3.4.2 Loadspace. The loadspace shall be defined as the full width of the platform and the length of the platform, from front to rear, ahead of the beavertail.

3.4.3 Loading diagram. The trailer manufacturer's recommended instructions for loading a forklift truck shall be inscribed on a plate permanently mounted on each side of the trailer. The plates shall conform to 3.1.1.11.

3.5 Dimensions and clearances. The trailer, without payload, uncoupled from the towing vehicle, resting on its landing leg on level ground, and with the platform level, shall conform to the dimensions specified in table I.

TABLE I. Dimensions and clearances.

	Designation	Dimensions	
	Overall length, mm (feet) (maximum)	6706	(22)
*	Overall width, size 6 and 9, mm (inches) (maximum)	2440	(96)
	Overall width, size 9-W, mm (inches) (maximum)	2590	(102)
	Platform floor height, mm (inches) (maximum)	560	(22)
	Beavertail floor height at rear, mm (inches) (maximum)	410	(16)
	Length of level loadspace (length of platform ahead of beavertail) mm (feet) (minimum)	3660	(12)
	Width of platform between wheelhousing and fenders; clear, unobstructed opening mm (inches) (minimum)	1880	(74)
	Beavertail and ramp angle, degrees	12 to 15	12 to 15
	Lunette height, measured from ground to centerline of lunette eye, vertical range of at least three positions, mm (inches) (minimum)	610 to 810	(24 to 32)
	Ground clearance to bottom of platform, mm (inches) (minimum)	300	(12)
	Ground clearance to bottom of beavertail at rear, mm (inches) (minimum)	250	(10)
	Vehicular ground clearance, exclusive of axles, mm (inches) (minimum)	230	(9)
	Vehicular ground clearance under axles and suspension, mm (inches) (minimum)	140	(5.5)

3.6 Frame and tow hitch. The trailer tow hitch and main frame members shall be adequately braced with a sufficient number of cross-members to ensure stability. When the trailer is loaded as specified in 3.4 and when being loaded, the maximum fiber stress in any frame, tow hitch, and platform member shall be not greater than 50 percent of the yield strength of the materials used. All parts of the frame shall be completely electric-arc

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welded. Where frame members are spliced, the joints shall be reinforced to fulfill the dynamic and static loading requirements specified herein. Splices shall be designed to avoid stress concentration. Splice welds shall be continuous.

3.6.1 Stress analysis. The contractor shall prepare load diagrams of the reactions at the lunette, landing gear leg, centerline of the bogie and the front and rear of the loading ramps, as applicable, when the trailer is empty, when in loaded travel position and when being loaded. The contractor shall determine and record sprung and unsprung weights. The longitudinal, vertical and transverse centers of gravity shall be determined for the unloaded and loaded conditions. Design load shall be equal to the product of 1.5 times the algebraic sum of the trailer curb weight minus the unsprung weight of the suspension system plus the rated payload distributed as specified in 3.4. The contractor shall prepare shear and moment diagrams with deflection and stress calculations, including buckling analysis for thin structural members. The maximum fiber stresses calculated shall be not greater than that specified in 3.6.

3.7 Axles. The axles shall be of a rated capacity at least equal to the loads imposed on each member, measured at the ground, with the trailer loaded as specified in 3.4.

* 3.8 Suspension system. The suspension system shall be of the full spring type. The suspension system shall have a rated capacity at least equal to the loads imposed with the trailer loaded as specified in 3.4. Clearances shall prevent interference between the wheels and tires and any portion of the vehicle under the articulation conditions specified for each size trailer.

3.8.1 Suspension for size 6 and size 9-W vehicles. Size 6, 5450 kg (6-ton) and size 9-W, 8150 kg (9-ton) trailers shall have tandem, through type axles mounting single wheels. The spring suspension shall provide sufficient articulation to maintain load contact of each wheel with the ground under each of the following conditions:

(a) With not more than two diagonally opposite wheels elevated 125 mm (5 inches);
(b) With the forward pair of wheels elevated 125 mm (5 inches) above the rear pair;
(c) With the rear pair of wheels elevated 125 mm (5 inches) above the forward pair.

3.8.2 Suspension for size 9 vehicle. Size 9, 8150 kg (9-ton) trailer shall have three through type axles mounting single wheels. The spring suspension shall provide sufficient articulation to maintain load contact of each wheel with the ground under each of the following conditions:

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(a) With the first and third wheels on one side and center wheel on the opposite side elevated 125 mm (5 inches) above the remaining wheels;
(b) With any pair of opposite wheels elevated 125 mm (5 inches) above the remaining wheels;
(c) With any two pairs of opposite wheels elevated 125 mm (5 inches) above the remaining pair.

3.9 Wheels, rims and tires. Rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Multi-piece rims shall not be furnished. Tire, rim size and load range (ply rating) shall be the same for all tires on each vehicle.

3.9.1 Tires. Steel belted radial tubeless truck type tires with highway tread shall be furnished. Tires shall be of a rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the trailer loaded as specified in 3.4. The number of tires furnished shall be four for size 6 and 9-W and six for size 9, plus any spares required. Tires and tire ratings shall conform to Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

* 3.9.2 Carrier for spare wheel assembly. A carrier for a spare wheel or rim and tire assembly shall be furnished. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the vehicle and carrier without personnel positioning themselves or any part of their body under the spare tire assembly. The carrier shall be installed in a readily accessible location on the vehicle.

3.9.3 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be provided. When a spare tire assembly is specified (see 3.9.4), a spare wheel or rim shall be furnished.

3.9.4 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design and load range (ply rating) as the tires furnished on the axles.

3.10 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.47. Brake linings shall be of nonasbestos material. Air or air-hydraulic brakes also shall conform to Federal Motor Carrier Safety Regulations 393.45 and 393.46. The brake system shall act on all wheels and shall have a rated torque capacity at least equal to the maximum torque imposed during braking, measured at the ground, with the trailer loaded as specified in 3.4. All components of the brake system shall be installed in such a manner as to provide adequate road clearance

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when traveling over uneven or rough terrain, including objects liable to strike and cause damage to the brake system components. No part of the brake system shall extend below the bottom of the wheel rims.

* 3.10.1 Service brakes (electric). Unless air brakes are specified (see 3.10.4), the trailer shall be furnished with a 12-volt dc electric brake system. The system shall include a manual controller kit for mounting on the dash of the towing vehicle and a controller kit for synchronizing the trailer electric brakes with the brakes of the towing vehicle. Unless otherwise specified, the synchronizing controller shall be for use with hydraulic brake equipped towing vehicles. When specified (see 6.2), the synchronizing controller shall be for use with air brake equipped towing vehicles. The kit shall be complete with all necessary parts and installation instructions. The kit shall be boxed, marked as to the contents and securely attached to the trailer for shipment.

3.10.2 Chock blocks. When electric service brakes are furnished, two pairs of wood, steel or plastic chock blocks with captive chains or cables and suitable cradles shall be installed on the trailer. The hook of each captive chain or cable shall be secured to the cradle. The captive chains or cables shall be long enough to be wrapped around the cradle to hold the chocks in place during travel and to allow the wheels to be chocked in both the front and the rear. The chock blocks shall be of sufficient size to prevent trailer movement during rear loading and unloading operations and while parked and headed both up and down a 30 percent slope.

3.10.3 Parking brakes (manual). When specified, (see 6.2), manual parking brakes shall be furnished when electric service brakes are furnished. The manual all-wheel parking brakes shall be designed to operate by mechanical actuation independent of the service system. Parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41. The parking brake lever shall have sufficient moment arm to require not more than 14 kg (30 pounds) of effort for application and release. When brake cables are used, they shall be corrosion-resistant.

3.10.4 Service brakes (air). When specified (see 6.2), an air brake system shall be furnished. The system shall use full-air-application internal expanding type brakes or air-over-hydraulic type brakes. The brake system shall include:

(a) Piping
(b) Hose connections
(c) Gladhands
(d) Spring loaded dust covers or dummy gladhands equipped with security chains or cables
(e) Provisions for stowing dummy couplings
(f) Air reservoirs with remote control drain located above the bottom of the frame rail
(g) Automatic slack adjusters for full air brakes

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(h) Relay emergency valve, and
(i) All other parts and connections required for a complete air-brake system.

Gladhands shall conform to SAE J318. The location of gladhands shall be as recommended in SAE J702. Slack adjusters and air chambers used shall be located above the bottom edge of the axle carrier. Hydraulic brake hose shall be in accordance with SAE J844.

3.10.5 Parking brakes (automatic). When full-air or air-over-hydraulic service brakes are furnished, automatic type parking brakes acting on the service brakes of all wheels shall be furnished. Parking brakes shall conform to Federal Motor Carrier Safety Regulation 393.41. Parking brakes shall be designed to operate in conjunction with the emergency and service brake air system. The brake actuating mechanism shall be capable of automatically applying the brakes both when the emergency air line pressure drops below normal operating pressure and when the trailer-to-truck air lines are disconnected. Brakes shall be capable of remaining in the applied position despite depletion of the compressed air supply. In the event of failure or depletion of the air supply, provisions shall be included to release the parking brakes manually without the use of tools.

3.11 Towing hitch. The towing hitch shall be of all-steel construction. The coupling device shall be a lunette ring of solid steel stock conforming to 3.1.1.10. The lunette shall have a vertical range of adjustment, measured at the centerline of the lunette, sufficient to cover a height range of not less than 610 mm (24) to not more than 810 mm (32 inches) above ground with at least three positions. The towing hitch and lunette shall conform to Federal Motor Carrier Safety Regulation 393.70. The towing hitch, and all its attachments, shall be capable of withstanding a 27,200 kg (60,000 pound) drawbar pull without residual deformation.

3.11.1 Safety chains. Safety chains shall be furnished. The chains shall conform to Federal Motor Carrier Safety Regulation 393.70(d) and to the method of attachment and capacity requirements of SAE J697. Cables shall not be furnished. A grab hook shall be provided on the free end of each chain.

3.12 Landing leg. A landing leg shall be mounted on the trailer drawbar frame. The landing leg shall have means for adjusting the leg vertically to enable personnel to couple and uncouple the trailer from a towing vehicle in any of the various ranges of height adjustment of the lunette. The landing leg shall be capable of raising and lowering its portion of the weight of the fully loaded trailer during the coupling and uncoupling operations without evidence of binding, damage, or deformation. The landing leg shall be retractable to provide adequate ground clearance for over-the-road travel. Means shall be provided to secure the landing leg in the travel position. Vertical adjustment shall be by means of a hand crank. The hand crank shall be secured to the vehicle with a corrosion-resistant chain or cable or shall

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be permanently attached to the landing leg. The leg adjusting mechanism shall be protected from exposure to foreign matter which would impair functional efficiency.

3.12.1 Landing leg ground plate or wheel. Unless otherwise specified, the landing leg shall be provided with a ground bearing plate with a minimum area of 129 square centimeters (cm^2) (20 square inches). When specified (see 6.2), the landing leg shall be provided with a caster wheel assembly with antifriction bearings.

3.13 Electrical system. All lights and reflectors shall be protected from operational hazards by mounting in recessed or otherwise guarded locations. Lights and reflectors shall not be mounted on vertical surfaces of the rub rail or on the trailer bumper. All electrical wiring shall conform to SAE J1292. Clearance and identification lamps shall be constructed for easy removal and replacement of lamps and lenses without the use of any tools.

3.13.1 Twelve-volt dc system. A 12-volt dc lighting system conforming to 3.13 through 3.13.1.2 shall be provided. The 12-volt dc lighting system shall conform to Federal Motor Carrier Safety Regulations 393.9, 393.11, 393.20, 393.22, 393.23, 393.25 through 393.29, 393.32 and 393.33.

3.13.1.1 Receptacle for 12-volt dc system. The front of the semitrailer shall be equipped with a 7-contact electrical receptacle conforming to SAE J560, with the connectors connected and color-coded as specified therein. The receptacle shall be provided with a spring loaded cover. The receptacle shall be located in accordance with SAE J702.

3.13.1.2 Seven-conductor 12-volt electrical cable. A seven-conductor electrical cable conforming to SAE J1067 shall be furnished. The cable shall be coiled and shall be not less than 910 mm (3 feet) longer than the distance from the 12-volt electrical receptacle to the lunette. Both ends of the cable shall be equipped with a round plug conforming to SAE J560. Each plug shall be equipped with a grip for withdrawing from the connector socket. Provisions shall be made to secure the cable, when not in use, on the drawbar or trailer body.

* 3.13.2 Brake wiring. Electric brakes shall be wired to the auxiliary circuit of the SAE J560 seven-conductor electrical receptacle. All wiring shall be properly mounted and secured with clamps. Brake wiring shall be heavy wall thermoplastic insulated cable and shall be enclosed in conduit.

* 3.13.3 Breakaway battery. When electric brakes are furnished, a 12-volt battery shall be installed to operate the electric brakes in the event of a breakaway from the towing vehicle. The battery shall be maintenance-free type having the maintenance-free characteristics listed in W-B-131. The reserve capacity rating measured in accordance with SAE J537 shall be not

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less than 35 minutes. The cold cranking rating at minus 18 degrees Celsius (°C) (0 degrees Fahrenheit (°F)) measured in accordance with SAE J537 shall be not less than 160 amperes. A weatherproof battery box shall be furnished.

* 3.13.4 Battery recharging circuit. A battery recharging circuit shall be wired through the first pin of the four-conductor electrical receptacle specified in 3.13.5. The second pin shall be return ground to the towing vehicle. A diode shall be installed in the system so that the electric current will not flow from the trailer battery to the towing vehicle system.

* 3.13.5 Breakaway battery wiring. The breakaway battery shall be wired to two conductors of a four-conductor electrical receptacle, Midland Model 23400 or equal. A second identical receptacle for mounting on a towing vehicle shall be furnished. The receptacles shall be polarized and weather-proof. A connecting cable, each end equipped with a mating plug, Midland Model 23403 or equal, shall be furnished. The cable shall be not less than 910 mm (3 feet) longer than the distance from the four-conductor receptacle to the lunette.

* 3.13.6 Winch battery. When an electric winch is furnished, a second battery to operate the winch shall be provided. The battery shall conform to 3.13.3, including a battery box.

* 3.13.7 Winch battery wiring. The battery for the winch shall be wired to the third and fourth pin of the four-conductor receptacle as specified in 3.13.4 and 3.13.5.

3.14 Fenders and wheelhousings. The trailer shall be equipped with heavy duty load bearing fenders. Fenders shall be capable of supporting at least a 115 kg (250 pound) load from a standing person applying hold-down chains to material handling equipment. The top surfaces shall be covered with safety steel tread plate of not less than 4.8 mm (0.188 inch) thickness, exclusive of the tread. Wheelhousings shall be constructed of sheet steel with sufficient cross-bracing and supports. The wheelhousings shall enclose the areas on the sides at least above the platform, and to the front and rear of the wheels. The wheelhousings shall prevent road splash onto the platform.

3.15 Rear wheel splash and stone throw protection. Mud flaps shall be provided to the rear of the rear wheels in accordance with SAE J682. A metal strip not less than 3.2 mm (0.125 inch) thick and not less than 25 mm (one inch) wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or the bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 25 mm (1 inch) high by 32 mm (1.25 inch) wide by 2.4 mm (0.094 inch) thick shall be furnished at each bolt.

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3.16 Rear end protection. Trailer rear end protection shall be in accordance with Federal Motor Carrier Safety Regulation 393.86.

3.17 Platform. The platform shall be of the level-deck type, with the minimum dimensions specified in table I.

* 3.17.1 Wood flooring. Unless steel grate flooring is specified (see 3.17.2), the platform shall be floored with hardwood conforming to 3.1.1.7 and 3.1.1.8. Wood flooring shall be laid lengthwise. Butt joints shall be over crossmembers and shall be staggered. Exposed ends and edges of flooring shall be protected with metal reinforcements. The floor shall be securely attached with corrosion-resistant countersunk bolts and bridge style floor clips to eliminate drilling frame members.

3.17.2 Metal grate flooring. When specified (see 6.2), the platform shall be floored with open bar type metal grating to allow passage of dirt, snow and sleet. The grating shall have serrated edges on the top surface to provide traction for forklift trucks and other material handling equipment in inclement weather. The underside of the grating shall be left open as much as practicable without undue reduction in the number or size of floor support members. The grating shall be securely attached to a supporting framework.

3.17.3 Curbing. Metal curbing, measuring not less than 115 mm (4.5 inches) in height above the platform, shall be provided along the full length of both sides of the platform and the beavertail. Curbing shall be provided at the front of the platform, not less than 200 mm (8 inches) in height above the floor level. Front curbing shall be fabricated of not less than a 200 mm (8-inch) channel. Side frame members and front crossmembers may be utilized to provide the required curbing or the curbing shall be welded to form an integral unit with the frame. The curbing shall withstand the impacts from a forklift truck of the gross weight specified herein or similar material handling equipment when winch-loaded or when driven aboard under its own power and impacting the curbing.

3.17.4 Beavertail. A beavertail shall be provided at the rear end of the platform. The beavertail shall have the same width and the same type of flooring and side curbing as the platform. The beavertail shall meet the dimensional requirements specified in table I.

3.17.5 Loading ramps. Two loading ramps shall be hinged to the rear of the beavertail. The width of each ramp shall be not less than 864 mm (34 inches). The ramp length shall be not less than 1220 mm (4 feet) and in no case less than the length required to provide the same slope as the beavertail, with the trailer platform level and the trailer and the rear of the ramps resting on level ground. The ramps shall be counter-balanced with torsion springs and shall be capable of being raised to an upright travel position; locked safely in the upright position for over-the-road travel; and unlocked and lowered gently to the ground; all by one man without the use of

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tools or auxiliary equipment. The ramps shall not infringe upon the platform loadspace when they are in travel position. The gap between the loading surface of the beavertail and the ramps shall be not greater than 25 mm (1 inch). The difference in elevation between the loading surface of the beavertail and the ramps at the hinged end shall be not greater than 6.4 mm (0.25 inch). The ramps shall provide adequate structural strength to withstand the wheel loadings encountered when loading forklift trucks and other material handling equipment of at least the gross weight specified herein. Ramp ground supports shall be integral with the ramps. The ramps may fold approximately midway of their length for over-the-road travel providing all the operational and structural strength requirements specified herein are met.

3.17.5.1 Ramp flooring. The rear (leading) edge of the ramps shall be surfaced with safety tread steel plate not less than 150 mm (6-inches) in width. The rear of the ramps shall have a wedge shaped cross-section permitting the top surface of the rear edges to rest not more than 13 mm (0.50 inch) above level ground. The entire top surface of the ramps shall be flat from front to rear. Unless otherwise specified (see 6.2), the ramps shall be floored the same as the platform and beavertail.

3.18 Tiedown D-rings. Tiedown D-rings of the heavy duty type shall be provided. D-rings shall not protrude beyond the overall width of the trailer when not in use. The trailer shall have five tiedown rings on each side of the platform, spaced on approximate 910 mm (36-inch) centers, with the front ring located near the front of the platform and the rear ring located just forward of the beavertail. Tiedown rings located along the frame adjustment to the fenders shall be of a type which are held in vertical position when not in use. Each tiedown ring and its attachments shall withstand a vertical load of not less than 2270 kg (5,000 pounds).

3.19 Winch. The trailer shall be equipped with a variable speed winch conforming to SAE J706. The winch shall be mounted forward of the platform and shall be capable of pulling a disabled forklift truck or similar wheeled vehicle up the ramp and beavertail incline and onto the loadspace. The winch shall be of the single drum type. An integral, adjustable, automatic safety brake shall be provided. The winch shall be provided with an anchor to retain the cable hook when the trailer is traveling.

* 3.19.1 Electric winch. Unless a hydraulic winch is specified (see 3.19.2), the winch shall be electrically operated. The electric winch shall have a single line pull capacity of not less than 3630 kg (8,000 pounds) on the bare drum. A wire rope not less than 23 meters (m) (75 feet) in length shall be furnished. The wire rope shall be not less than 8.0 mm (0.313 inch), aircraft cable with end chain and hook.

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3.19.2 Hydraulic winch. When specified (see 6.2), the winch shall be hydraulically operated. The hydraulic winch shall have a single line pull capacity of not less than 9070 kg (20,000 pounds) on the bare drum. The winch shall be provided with a hydraulic relief valve set to restrict the line pull to the winch rated capacity. Wire rope for the hydraulic winch shall be not less than 14 mm (0.563 inch), aircraft cable, with end chain and hook. The cable shall be not less than 23 m (75 feet) in length.

3.19.3 Hydraulic system. When a hydraulic winch is specified, a self-contained hydraulic system shall be installed. The hydraulic system shall consist of:

(a) An air cooled diesel engine;
(b) A hydraulic pump;
(c) Hydraulic cylinders;
(d) A hydraulic fluid reservoir;
(e) Control and relief valves;
(f) Filters; hoses;
(g) Tubing; fittings; and
(h) All other components required to actuate and control the hydraulic powered winch.

3.19.3.1 Hydraulic pump. The hydraulic pump shall be capable of delivering a flow rate at maximum system operating pressure which is adequate to provide a winch line speed of 1.5 m per minute (5 feet per minute) under all operating load conditions.

3.19.3.2 Hydraulic reservoir. A hydraulic reservoir having sufficient capacity to hold the maximum volume of fluid required for the system plus a minimum 20 percent reserve shall be provided. The reservoir shall be vented by a separate vent or through the filler cap. The vent shall be fitted with a micronic filter element.

3.19.3.3 Hydraulic lines and fittings. All hydraulic tubing, hoses and fittings shall conform to applicable SAE standards. A minimum number of fittings, joints and connections shall be used to prevent excessive back pressure, clogging and leakage. All hoses and fittings shall be capable of withstanding a bursting pressure of four times the working pressure and a maximum proof-pressure of at least two times the working pressure. Hydraulic hoses shall be single wire braid or double wire braid, rubber covered, conforming to SAE J517. Hose fittings shall conform to SAE J516. The hydraulic system shall incorporate a filtration system conforming to SAE J931. Hoses shall be protected against chafing through utilization of grommets, supports and protective devices.

3.19.3.4 Hydraulic fluid and filter. A full-flow hydraulic filter, having a micron rating not greater than 35, shall be installed in the hydraulic system. The hydraulic fluid shall be of a noncorrosive type with

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characteristics to withstand a temperature range from -40°C (-40°F) to +135°C (+275°F). When specified (see 6.2), hydraulic fluid conforming to MIL-H-46170 shall be provided in the hydraulic system. A plate shall be located near the hydraulic system fill port, stating "CAUTION: USE HYDRAULIC FLUID CONFORMING TO MIL-H-46170 ONLY." The plate shall conform to 3.1.1.11.

3.19.4 Diesel engine. The hydraulic pump shall be driven by an air cooled diesel engine. The engine power, speed and torque characteristics shall be sufficient to provide continuous operation of the hydraulic system at the flow rate specified in 3.19.4.1 and the pressure required to operate the winch under maximum load. The engine shall be provided with an ether starting system, glow plug or electric grid heater. If an ether system is furnished in lieu of a glow plug, it shall be key operated or manually operated, and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 355 milliliters (mL) (12 fluid ounces) shall be furnished. A reservoir need not be furnished.

* 3.19.4.1 Sound level. The noise level during winch operation shall be not more than 84 db(A) when measured in accordance with MIL-STD-1474. The noise level shall be measured at the operator's station adjacent to the mounted winch controls, with the operator standing on the ground.

3.20 Servicing and adjusting. Prior to acceptance by the Government inspector, the contractor shall service and adjust each trailer for operational use including at least the following:

(a) Inflation of all tires
(b) Adjustment of brakes
(c) A check for proper functioning of all lights
(d) Filling and charging of batteries
(e) Complete lubrication with grades of lubricants recommended for the ambient air temperature at the delivery point.

3.21 Workmanship. Defective components, parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Bolted, riveted and welded construction used shall be in accordance with the highest standards of the industry.

* 3.21.1 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to ensure uniformity of size and shape.

* 3.21.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

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1.1 the power supply shall

* 3.21.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

* 3.21.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. See 3.1.1.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

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4.3 First production vehicle inspection. The first vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.3.1 Vehicle weight. The first production trailer, fully equipped, shall be weighed to determine the curb weight and distribution of the curb weight on the lunette and on the axles. The total imposed loading on the lunette and axles with the trailer loaded as specified in 3.4 shall be computed by the contractor and verified by the Government using the curb weight and the payload. The calculated imposed loads on the lunette and the axles shall be compared to the suspension, axle and tire load capacity ratings to determine if these components are of adequate capacity to meet contractual requirements.

* 4.3.2 Road test. The first vehicle for each contract line item number (CLIN) under contract shall be road tested with and without payload. The road test shall be for the distance required in table II, corresponding to the number of vehicles being procured under the specific contract line item number (CLIN). The test shall be conducted by the contractor under the direction of Government representatives. The test shall be conducted over a test course approved by the procuring activity and meeting the requirements of table III. The vehicle shall be inspected periodically during and at the completion of the test by the contractor.

TABLE II. Road test distance.

Number of vehicles being procured	Road test distance km (miles)
1 - 10	160 (100)
11 - 20	320 (200)
21 - 30	480 (300)
31 - 40	640 (400)
41 - 50	800 (500)
51 - 100	1600 (1000)
101 - up	8050 (5000)

TABLE III. Test course requirements.

Type of road	Road test distance
Dirt	10 percent
Gravel	30 percent
Improved pavement	60 percent

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* 4.3.2.1 Test course. The total test mileage shall be on roads and highways within the general geographic limits of the place of manufacture and adjoining states. Any special highway permits required shall be obtained by the contractor. The course shall be arranged to return the vehicle to the manufacturer's plant at the completion of each 1600 km (1,000 miles), if applicable, or at the completion of the test otherwise, for inspection and chassis servicing.

* 4.3.2.2 Test loads. The trailer shall be tested for 50 percent of the mileage with the trailer loaded as specified in 3.4 and for 50 percent of the mileage with the trailer empty. Average speeds during all road tests shall be as close as possible to those specified in 3.3.

* 4.3.2.3 Test conditions. At least five sudden stops shall be made from a speed of not less than 32 km/h (20 mph). Tracking ability shall be observed to verify conformance to 3.3.3; turning ability shall be observed to verify conformance to 3.3.1. After the test, the trailer shall be examined for evidence of misalinement, binding or other malfunction.

4.3.2.4 Electrical test. An electrical test shall be conducted on the first production vehicle to determine that all vehicle lights are functioning. In addition, when an interconnected 24-volt system is furnished the circuits of the 12-contact receptacle shall be energized with not less than 28 volts dc regulated voltage for a time sufficient to stabilize the temperature of the resistor assembly for not less than 30 minutes in an ambient temperature of not less than 25°c (77°F). After the test, the resistor assembly, including resistors, wiring and adjacent components, shall be examined for evidence of overheating, deterioration or damage.

4.3.2.5 Wood treatment certification. Manufacturer's records, acceptable to the Government, shall be available for verification that all wood requiring treatment in accordance with MIL-STD-1223 has been treated.

4.3.3 Production sample. Upon acceptance of the first production trailer, it shall remain at the manufacturing facility as a production sample, and be the last trailer shipped on the contract. The supplier shall maintain the vehicle in an as new condition for the duration of the contract. Release of the first production vehicle for shipment prior to completion of the contract shall be at the discretion of the local Government Quality Assurance Representative (QAR).

4.3.3.1 Delivery of test vehicle. Upon successful completion of the vehicle test and final examination, the vehicle shall be cleaned, serviced and refurbished to the extent required for delivery of a new vehicle to the Government. Warranty provisions of the contract shall be exclusive of total test miles. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

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4.3.4 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of vehicle until corrective action has been taken. In addition, subsequent vehicles that do not conform to the accepted production sample vehicle shall be rejected until corrective action has been taken. In the event of a failure, any or all previously accomplished tests may require repetition at the discretion of the Government.

4.4 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting have been accomplished.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The trailers covered by this specification are intended for general use by the Government in retrieving forklift trucks and similar wheeled material handling equipment and transporting them over highways, improved and unimproved roads.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Size of trailer required (see 1.2).
(c) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Identification of appropriate military service for treatment, painting and marking (see 3.1.1.1 and 3.1.1.2).
(e) Independent 24-volt dc system, if required (see 3.1.1.9.1).
(f) Interconnected 24-volt dc system, if required (see 3.1.1.9.2).
(g) Spare wheel or rim, if required (see 3.9.3).
(h) Spare tire assembly, if required (see 3.9.4).
* (i) Towing vehicle synchronizing controller for use with air brake towing vehicles, if required in lieu of synchronizing controller for use with hydraulic brakes (see 3.10.1).
(j) Manual parking brakes, if required (see 3.10.3).
(k) Air brake system in lieu of electric brakes, if required (see 3.10.4).

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(l) Support leg caster wheel assembly in lieu of ground plate, if required (see 3.12.1).
(m) Steel grate flooring in lieu of wood, if required (see 3.17.2).
(n) Ramp flooring requirement, if not the same as the platform (see 3.17.5.1).
(o) Engine driven hydraulic winch in lieu of electric, if required (see 3.19.2).
(p) Hydraulic fluid conforming to MIL-H-46170, if required (see 3.19.3.4).

6.3 Subject term (key word) listing.
Electric winch
Hydraulic winch
Lashing ring
Loading ramp
Non-tactical vehicle (NTV)
Trailer, commercial
Welder's qualifications.

6.4 Identification of changes. The margins of this document are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, regardless of the marginal notations and relationship to the previous issue.

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Air Force - 99

Preparing Activity:
Army - AT

Project No. 2330-A135

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-62109C	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE TRAILERS, LOWBED: FORKLIFT TRANSPORTER, 5450 AND 8150 KILOGRAM (6 AND 9 TON), MODIFIED COMMERCIAL

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial: (313)754-6407 (2) AUTOVON: 786-6407
c. ADDRESS (Include Zip Code) US Army Tank Automotive Command Warren, Michigan 48397-5000 AMSTA-UED	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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